



QuantumNet: An enhanced diabetic retinopathy detection model using classical deep learning-quantum transfer learning[☆]

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ABSTRACT

Diabetic Retinopathy (DR), a diabetes-related eye condition, damages retinal blood vessels and can lead to vision loss if undetected early. Precise diagnosis is challenging due to subtle, varied symptoms. While classical deep learning (DL) models like CNNs and ResNet's are widely used, they face resource and accuracy limitations. Quantum computing, leveraging quantum mechanics, offers revolutionary potential for faster problem-solving across fields like cryptography, optimization, and medicine. This research introduces QuantumNet, a hybrid model combining classical DL and quantum transfer learning to enhance DR detection. QuantumNet demonstrates high accuracy and resource efficiency, providing a transformative solution for DR detection and broader medical imaging applications. The method is as follows:

- Evaluate three classical deep learning models—CNN, ResNet50, and MobileNetV2—using the APTOS 2019 blindness detection dataset on Kaggle to identify the best-performing model for integration.
- QuantumNet combines the best-performing classical DL model for feature extraction with a variational quantum classifier, leveraging quantum transfer learning for enhanced diagnostics, validated statistically and on Google Cirq using standard metrics.
- QuantumNet achieves 94.11 % accuracy, surpassing classical DL models and prior research by 11.93 percentage points, demonstrating its potential for accurate, efficient DR detection and broader medical imaging applications.

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Background

Diabetic Retinopathy (DR) is a significant complication of diabetes, characterized by damage to the retinal blood vessels, potentially leading to vision loss or blindness if untreated. Early detection and diagnosis are critical for effective management. The growing prevalence of diabetes globally, particularly in regions like Southeast Asia where diabetic cases are projected to rise from 11.3 % in 2019 to 12.2 % by 2030, underscores the urgent need for scalable and efficient diagnostic solutions [1,2].

The manual diagnosis of DR poses challenges due to the need for highly specialized expertise to detect subtle retinal abnormalities such as microaneurysms, hemorrhages, and exudates. Automated diagnostic tools powered by artificial intelligence (AI) and machine learning (ML) offer promising solutions. Traditional deep learning (DL) models such as Convolutional Neural Networks (CNNs) and variants like ResNet and MobileNet have demonstrated considerable success in medical imaging tasks. However, these models often face limitations in terms of computational efficiency, generalization, and sensitivity to complex medical datasets. The potential of quantum computing to enhance computational speed and accuracy motivates the exploration of hybrid approaches integrating classical DL and quantum techniques [3,4].

While classical DL models have achieved commendable performance in DR detection, they remain constrained by their reliance on resource-intensive training processes and limitations in capturing the intricate patterns inherent in medical imaging. Quantum computing, leveraging principles such as superposition and entanglement, offers a pathway to overcome these challenges. The key problem addressed in this study is the development of a hybrid model that combines the strengths of classical DL models for feature extraction with quantum transfer learning for enhanced classification performance. This approach aims to improve diagnostic accuracy while maintaining computational efficiency [5,6]. Therefore, the main objectives of this study are:

- **Evaluate Classical DL Models:** Assess the performance of CNN, ResNet, and MobileNet architectures in DR detection using the APTOS 2019 blindness detection dataset. These models serve as baselines for feature extraction and classification capabilities [7,8].
- **Develop a Hybrid Model - QuantumNet:** Design and implement QuantumNet, a hybrid framework that integrates the best-performing classical DL model (ResNet-50) with a variational quantum classifier (VQC). The hybrid approach leverages classical DL for robust feature extraction and quantum computing for enhanced classification [9,10].
- **Validate Performance Gains:** Demonstrate the superiority of QuantumNet over classical models through quantitative metrics such as accuracy, precision, recall, and F1-score, validated statistically using techniques like the Wilcoxon rank test [11,12]. The comparative evaluation of these models allows us to quantify the performance gains of advanced architectures (ResNet) over traditional methods (CNN), assess the feasibility of deploying resource-efficient models (MobileNet) in practical environments and establish a strong rationale for integrating hybrid quantum-classical models, such as QuantumNet, which leverage the strengths of ResNet for feature extraction while addressing computational limitations through quantum transfer learning.
- **Address Practical Challenges:** Evaluate the practical challenges associated with implementing hybrid quantum-classical models, including hardware constraints, error rates in quantum systems, and scalability for real-world deployment and propose strategies for overcoming these limitations [13,14]. This study represents a pioneering step toward leveraging quantum computing in medical imaging, specifically for diabetic retinopathy detection. By addressing both computational and diagnostic challenges, QuantumNet exemplifies the potential of hybrid AI-quantum frameworks to revolutionize healthcare diagnostics. The findings of this research have implications for broader medical imaging applications, setting the stage for future advancements in hybrid quantum-classical AI systems [15–19].

Method details

This section outlines the dataset overview, pre-processing techniques, model architecture, methodology and the evaluation metrics used.

Dataset overview

The dataset used originates from a publicly accessible repository of retinal fundus images available on Kaggle, called the APTOS 2019 Blindness Detection dataset [20]. It consists of 3662 high-resolution (584×565 pixels) fundus images in total collected from multiple medical institutions, representing a diverse demographic and geographic patient population. These images were captured by medical technicians to label the blood vessels for ophthalmology assessment and diagnosis, a procedure known to be labor-intensive and resource demanding. DR severity is used to classify the input images, which are then divided into several groups for a total of 3662 images as mentioned below:

- **No DR (Class 0):** Images showing no signs of DR. These total 1805 images.
- **Mild DR (Class 1):** Minimal microaneurysms but no significant damage. These total 370 images.
- **Moderate DR (Class 2):** Some hemorrhages and microvascular abnormalities, but no significant vision threat. These total 999 images.
- **Severe DR (Class 3):** Numerous hemorrhages and microvascular changes with a risk of vision loss. These total 193 images.
- **Proliferative DR (Class 4):** Advanced disease with new blood vessel growth, scar tissue, and a high risk of vision loss and blindness. These total 295 images.

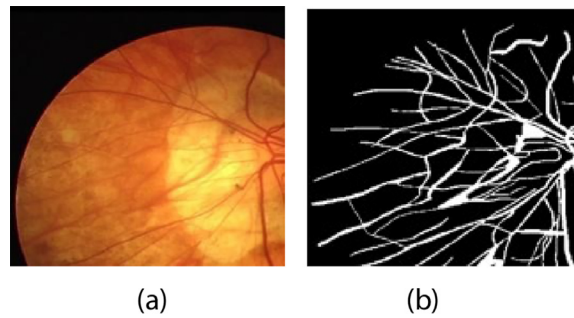


Fig. 1. Illustration of (a) a training set image; and (b) its corresponding label. In the label, the blood vessels are depicted as white, while the retinal background is represented in black.

Fig. 1(a) and (b) illustrate an example image from the dataset along with its corresponding binary mask label. Hence, in the proposed study, this dataset serves the purpose of enabling computer-aided screening of images by performing binary semantic segmentation, where each pixel is classified as belonging to a blood vessel or not without direct ophthalmologist intervention, facilitating timely disease detection. The focus in this study is on binary classification task for specific analysis (e.g., DR vs. non-DR), which is a common approach in medical imaging studies, where detecting the presence of a condition (DR) is often prioritized over grading its severity. Clinically too, the most critical question is whether a patient requires treatment (DR vs. non-DR), rather than the exact severity level, especially in resource-limited environments. Also, if the model performs well in binary classification, its adaptability to multi-class tasks can be better justified.

Image pre-processing

The retinal images are pre-processed to improve their quality and enable efficient model learning before training the standalone DL and hybrid QuantumNet model. The steps followed are:

- **Rescaling:** The images were rescaled by dividing each pixel value by 255.0 to normalize the pixel intensities between 0 and 1. Dividing by 255.0 ensures that the operation is performed using floating-point arithmetic, maintaining precision and preventing truncation errors that might arise with integer division in some programming environments. And normalizing input data helps the model converge faster and improves numerical stability during training.
- **Image Augmentation:** Various image augmentation techniques e.g. rotation, width, height shifts, shear, zoom, and horizontal flip are applied. By using these methods, the training data becomes more diverse and the model's capacity for generalization is enhanced.

Following data preprocessing, a refined dataset is curated and categorized into two groups. For model training, 70 % of data was set aside out of which 1293 images without DR (non-DR) and 1270 images depicting DR were employed. Model validation at a 15 % split utilizes 549 images, comprising 275 non-DR and 274 DR images. The next step in evaluating the performance of the model was to test 549 randomly chosen images, which is balance 15 % of the data on both hybrid DL-QTL-based and conventional DL models. The rationale behind the 70 %–15 %–15 % split is that it is a well-established and widely adopted strategy in ML for datasets of moderate size, such as the APTOS 2019. The larger 70 % training set helps improve the model's generalizability by exposing it to diverse examples and for it to learn the patterns and features from the dataset effectively. The 15 % validation set is used to tune hyperparameters, monitor model performance to unseen data during training phase, and avoid overfitting. The balance 15 % test set is exclusively reserved for the final unbiased evaluation of the model's performance after training is complete. Overall, the study's primary objective is to demonstrate the superiority of QuantumNet over classical models. The 70 %–15 %–15 % split allowed focused validation of QuantumNet's performance using independent test data, ensuring unbiased results.

Flow from Directory: The data was loaded using the '*flow_from_directory*' function, which generates batches of augmented images and their corresponding labels from the specified directory. This function facilitates efficient loading and preprocessing of large image datasets.

Model architecture

The models used in this study are briefly discussed in this section.

- **Convolution Neural Network (CNN)** — CNN specializes at processing data with a grid-like structure, such as images. The CNN model employs a fully connected layer structure as a classifier after feature extraction, as illustrated in Fig. 2 [21]. This includes flattening the output of the final convolutional layer, one or more dense layers that reduce the feature dimensions and a Softmax activation layer for classification into DR severity classes. A high degree of accuracy can be achieved by using fully connected layers for classification [22,23].
- **MobileNet** — MobileNetv2 is tailored for mobile applications integrating a lightweight classification head, that features a global average pooling layer followed by a fully connected layer optimized for efficiency. It provides residual connections between

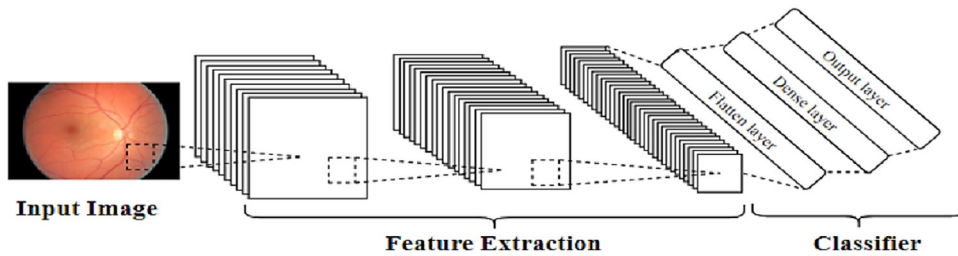


Fig. 2. CNN architecture for DR detection.

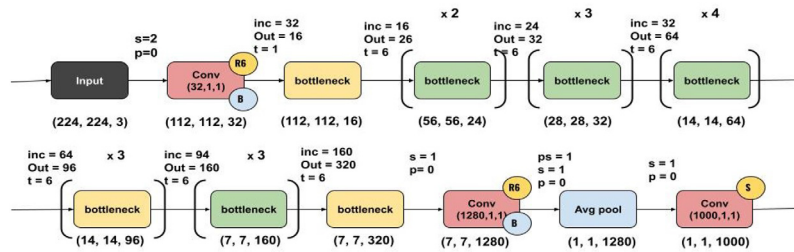


Fig. 3. MobileNet architecture for DR detection.

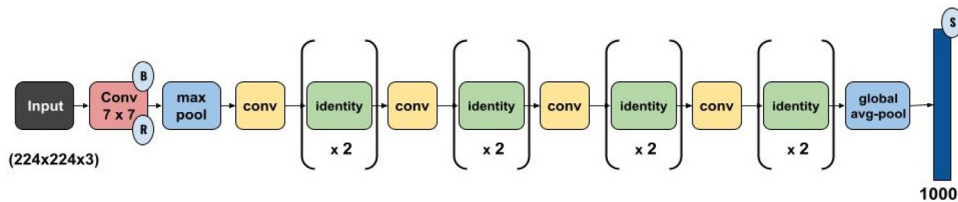


Fig. 4. ResNet architecture for DR detection.

bottleneck layers by using an inverse residual structure. Non-linearity is introduced by intermediate expansion layer filters using lightweight depth-wise convolutions. MobileNetv2 incorporates 19 residual bottleneck layers and a fully convolutional layer with 32 filters. A softmax activation function maps the feature vectors to class probabilities. A high-level schematic of the MobileNet architecture used for DR diagnosis is shown in Fig. 3.

- **ResNet** — ResNet-50 is a convolutional neural network comprising 50 layers designed for DL tasks [24,25]. The system uses a pre-trained version from the extensive ImageNet dataset which has been trained on over a million images. Consequently, the network boasts a repository of intricate feature representations spanning a broad spectrum of images. With an image input resolution of 224×224 pixels, the model's architecture is illustrated in Fig. 4. It uses a global average pooling layer to reduce the feature map size before passing it to the fully connected layer(s). This reduces the risk of overfitting by minimizing the number of trainable parameters. The final layer applies softmax activation for classification.

The choice of ResNet50 over other ResNet versions was to strike an optimal balance between depth (50 layers) and computational cost. While deeper variants like ResNet101 or ResNet152 can extract more complex features, they also require significantly more computational resources and are prone to overfitting on smaller datasets. Compared to shallower versions like ResNet18 or ResNet34, ResNet50 provides greater representational capacity, enabling it to capture subtle features necessary for complex tasks such as diabetic retinopathy detection.

The retinal images, after preprocessing, are fed into their corresponding DL models for training purposes. Utilizing the APTOS 2019 dataset, the models undergo training, with ground truth annotations acting as the target labels. To assess the models' performance, a distinct test set is employed, consisting of retinal images unseen during training. Subsequently, the trained models predict outcomes for these unseen images, and their performance is gauged using standard evaluation metrics.

- **QuantumNet** — QuantumNet is designed as a two-part system where the first part uses a classical deep learning model for feature extraction, and the second part employs a quantum transfer learning model for classification.

Classical deep learning model

A pre-trained CNN, specifically ResNet-50 is used due to its robustness in extracting hierarchical features from medical images. The choice of ResNet-50 allows us to leverage its deep residual layers, which are highly effective in capturing subtle features in retinal

images that are crucial for identifying various stages of DR. To reduce dimensionality and prepare the data for quantum processing, a global average pooling layer is applied to the output of ResNet-50, specifically to the feature maps from the last convolutional layer.

Quantum computing preliminaries

Quantum computing uses the principles of quantum mechanics to perform computations. The key quantum concepts used in this study are:

Variational Quantum Circuits (VQC): A Variational Quantum Circuit (VQC) is a hybrid quantum-classical algorithm framework designed for solving optimization and machine learning problems. It uses a parameterized quantum circuit and a classical optimization loop. Its key features include:

- **Parameterized gates:** The circuit contains adjustable parameters (like angles in rotation gates) that are optimized during the training process.
- **Cost function:** A classical computer evaluates a cost function based on the output of the quantum circuit.
- **Optimization loop:** Classical optimization algorithms (e.g., gradient descent) adjust the parameters to minimize the cost function iteratively.

Hadamard Gates: The Hadamard gate (denoted H) is a single-qubit gate that creates superposition. It transforms a qubit into an equal superposition of $|0\rangle$ and $|1\rangle$. It operates as:

- **Input: $|0\rangle$**

$$H | 0 \rangle = | 0 \rangle + | 1 \rangle / \sqrt{2}$$

- **Input: $|1\rangle$**

$$H | 1 \rangle = | 0 \rangle - | 1 \rangle / \sqrt{2}$$

It is represented in Matrix form as:

$$H = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

It is used to create superposition. It forms part of quantum algorithms like the Quantum Fourier Transform and Grover's algorithm.

$R_y(\theta)$ gates: It is a single-qubit rotation gate that rotates a qubit around the y-axis of the Bloch sphere. It operates by applying a rotation by an angle θ around the y-axis. It is represented by:

$$R_y(\theta) = \begin{vmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{vmatrix}$$

Parameterized R_y gates are used to tune the state of qubits based on optimization parameters in Variational Quantum Circuits.

$U(\alpha, \beta, \gamma)$ circuit: It refers to a unitary quantum gate parametrized by angles (α, β, γ) that defines the rotation of a qubit state in the quantum state space used in quantum circuits to perform operations on qubits. It represents a general-purpose parameterized quantum gate, consisting of a combination of rotation gates and phase shift gates to achieve a comprehensive transformation of the quantum state. It is used in variational quantum circuits (VQC) for encoding and manipulating quantum states.

CNOT Gates: The Controlled-NOT (CNOT) gate is a two-qubit quantum gate that flips (applies a NOT operation to) the target qubit if the control qubit is in the $|1\rangle$ state. It introduces quantum entanglement between qubits. It is a critical building block for multi-qubit quantum algorithms and variational quantum circuits.

Quantum computing model

The variational quantum classifier (VQC) is employed which consists of a parameterized quantum circuit. This circuit is designed to operate on a hybrid quantum-classical interface, where classical data is encoded into quantum states through an angle embedding strategy. The encoded quantum data is then processed through a series of quantum gates that are variably adjusted during training to optimize the model's parameters. Specifically, it uses a 4-qubit "dressed quantum circuit" to classify the feature vector from a 2048-dimensional space to 4 dimensions, as shown in Fig. 5. Quantum gradient descent, a method assembled especially for dealing with non-convex optimization landscapes seen in quantum circuits, is used in the optimization process.

The integration of classical deep learning with quantum computing in QuantumNet involves a seamless flow where the extracted features from ResNet are converted into a format suitable for quantum processing. This process at a high level involves the following steps:

Data Encoding: Classical data from ResNet is encoded into quantum states using an angle embedding technique, where each feature is mapped to a rotation angle of a quantum gate, thereby preparing the quantum circuit with the data-specific initial state. The data encoding steps include:

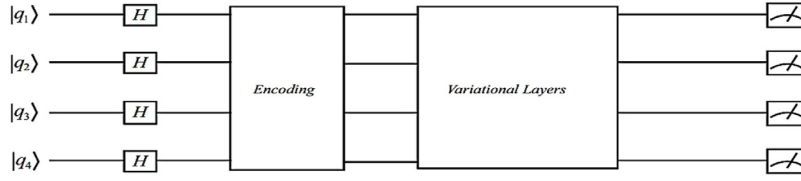


Fig. 5. Variational quantum classifier of 4-qubits for DR detection.

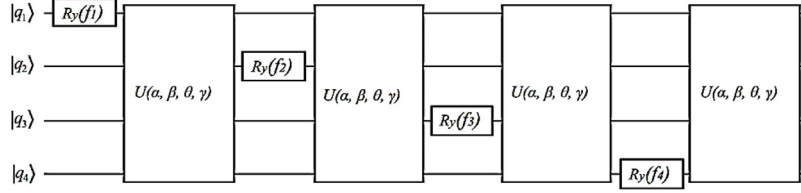


Fig. 6. Data encoding circuit.

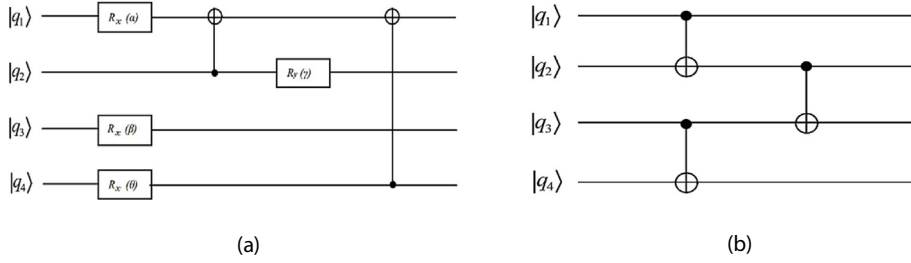


Fig. 7. (a) $U(\alpha, \beta, \theta, \gamma)$ circuit; (b) Entanglement layer circuit.

Initializing four qubits in the $|0\rangle$ state. Applying Hadamard (H) gates to induce superposition, rendering them in states of zero or one.

Using a unitary circuit, more modifications are employed to encode classical data. In this procedure, the data encoding circuit, which consists of $R_y(f_i)$ gates, encodes the 4-dimensional feature vector as weights and the $U(\alpha, \beta, \theta, \gamma)$ circuit, as illustrated in Fig. 6 and represented by Eq. (1).

$$\epsilon(x) = \bigotimes_{k=1}^4 \left(R_y \left(x_k \frac{\pi}{2} \right) |0000\rangle \right) \quad (1)$$

Here, $U(\alpha, \beta, \theta, \gamma)$ encodes classical data into quantum states through parameterized rotations. The gate acts as a building block for variational layers in the quantum circuit, enabling tunable operations during model training. The parameters $(\alpha, \beta, \theta, \gamma)$ are optimized during training to learn the optimal quantum state transformations required for classification. And Eq. (1) describes the ‘Data Encoding’ step on how classical data is encoded into quantum states using an angle embedding strategy. Each feature in the input data is mapped to the rotation angle of a quantum gate, specifically the gate, which ensures that classical features are effectively represented in the quantum domain.

As shown in Fig. 7, a trainable sequence made up of variational layers that include a data encoding circuit along with an entanglement layer, comprises three CNOT gates entangling all qubits [26].

$$Q = L_6 \circ L_5 \circ L_4 \circ L_3 \circ L_2 \circ L_1 \quad (2)$$

Eq. (2) outlines the structure of variational layers, where parameterized quantum gates interact with the encoded data. The layers include entanglement operations (e.g., CNOT gates) to enable qubit interactions, enhancing the model’s ability to learn complex relationships.

$$L(w) : |x\rangle \rightarrow |y \geq k \bigotimes_{k=1}^4 R_y(w_k) |x\rangle \quad (3)$$

Eq. (3) elaborates on how the variational parameters are optimized during training using hybrid classical-quantum algorithms. It emphasizes the iterative adjustment of these parameters to minimize the loss function, improving the classification accuracy.

$$k = (\text{CNOT} \otimes ||_{3,4}) (||_{1,2} \otimes \text{CNOT}) (||_1 \otimes \text{CNOT} \otimes ||_4) \quad (4)$$

Eq. (4) describes the measurement step, where the quantum circuit’s output states are collapsed into classical values. The probability of measuring a specific outcome is linked to the model’s predictions.

Finally, the measurement procedure is completed to determine the 4-qubit's predicted value about the Z operator as shown in Eq. (5). It formalizes the relationship between the measured quantum state and the final prediction. It involves aggregating the results of multiple measurements to compute the final class probabilities, ensuring reliable decision-making.

$$\begin{aligned}
 M(|y\rangle) &= y = \langle y | Z \otimes I \otimes I \otimes I | y \rangle \\
 &= \langle y | I \otimes Z \otimes I \otimes I | y \rangle \\
 &= \langle y | Z \otimes I \otimes Z \otimes I | y \rangle \\
 &= \langle y | Z \otimes I \otimes I \otimes Z | y \rangle
 \end{aligned} \tag{5}$$

Hybrid Training: During the training phase, a hybrid optimizer that leverages both classical backpropagation for the CNN components and quantum gradient descent for the quantum circuit is employed. This allows for simultaneous tuning of both classical and quantum parameters, enhancing the model's ability to generalize across different stages of DR more effectively.

Classical backpropagation for CNN components differs from quantum gradient descent for the quantum circuit as below. Classical backpropagation in CNNs operates on numerical data using deterministic mathematical operations like matrix multiplications and chain rule-based gradient calculations to optimize parameters. It updates these using classical optimizers such as SGD or Adam. In contrast, quantum gradient descent works within quantum systems, manipulating quantum states encoded in qubits via unitary transformations. Gradients are calculated using quantum-specific methods like the parameter-shift rule. While classical systems are robust and scale well with mature hardware, quantum systems face challenges like noise, decoherence, and hardware limitations. Despite their shared goal of minimizing a loss function, their methodologies differ due to their distinct computational paradigms.

Performance & Validation: The classical DL models and QuantumNet are trained and validated on the APTOS 2019 blindness detection dataset using the standard evaluation metrics.

Quantum device

The Google Cirq simulator [26], a framework developed specifically for designing, simulating, and testing quantum circuits is used as the quantum device. The reason for its choice is because Cirq provides a controlled environment for simulating the behaviour of quantum circuits without requiring physical quantum hardware. This is particularly beneficial in the current era where quantum devices are limited by noise and error rates. The simulator enables efficient prototyping and testing of variational quantum circuits (VQCs). Researchers can design, optimize, and test circuits for hybrid quantum-classical machine learning tasks before transitioning to actual quantum devices. Also, Cirq's user-friendly interface and extensive library of tools allows to integrate quantum simulations seamlessly with classical machine learning workflows, as demonstrated in QuantumNet. By leveraging Cirq, the study ensures that the quantum components of QuantumNet are thoroughly validated under idealized conditions. This enables reliable benchmarking of hybrid quantum-classical approaches against purely classical deep learning methods. While the simulator offers several advantages, it is not without limitations and challenges, which have been carefully considered in this research work:

- One of the most significant limitations of using a simulator like Cirq is that it does not perfectly mimic the behavior of real quantum hardware. Simulators can run idealized computations that may overlook practical challenges such as quantum decoherence and operational errors which are prevalent in actual quantum devices.
- Although simulators function exceptionally well for small to medium-sized quantum circuits, their performance degrades as system size increases to simulate larger quantum systems.

The research acknowledges that while the simulator offers a practical and accessible means to explore and develop quantum-enhanced machine learning models, the transition to actual quantum hardware is necessary to fully realize and validate the potential advantages of quantum computing in practical applications, such as enhanced computational speed and efficiency.

Evaluation metrics

To evaluate the models' performance, standard metrics i.e., model accuracy, precision, recall, and F1-Score are considered. While one model may exhibit high accuracy, it might not necessarily perform well in terms of recall, which is critical for ensuring that fewer cases of DR go undetected. Accordingly, Precision helps to understand the accuracy of positive predictions, which is crucial for medical diagnostics where the cost of a false positive can be high. Recall measures the ability of a model to find all the relevant cases (positive samples), an essential factor in medical screening tests. The F1-score helps in balancing the trade-offs between precision and recall, particularly important in our context where both false positives and false negatives carry significant importance. We use the above performance metrics to plot the Confusion Matrix for all models. These matrices summarize the classification performance for Diabetic Retinopathy (DR) and Non-Diabetic Retinopathy (Non-DR) categories from the validation dataset which contains 549 images, split as DR (274 images) and Non-DR (275 images). Each matrix includes TP: True Positives (Correct DR Predictions), FN: False Negatives (Missed DR Cases), FP: False Positives (Incorrect DR Predictions), and TN: True Negatives (Correct Non-DR Predictions).

Additionally, statistical analysis is performed to determine the significant performance differences observed between classical DL models and the hybrid QuantumNet model. The Wilcoxon rank test, which is a non-parametric test is used. The motivation for choosing this test firstly is that it is a robust, non-parametric statistical evaluation. Second it provides a reliable comparison of paired performance metrics (e.g., accuracy) of QuantumNet against the classical models (CNN, ResNet, and MobileNet). And lastly there is minimized influence of outliers and assumptions about data distribution. The test evaluates whether the observed differences in

Table 1
Model architecture of classical DL models.

(a) CNN		
Layer (Type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
flatten (Flatten)	(None, 186,624)	0
dense (Dense)	(None, 256)	47,776,000
dense_1 (Dense)	(None, 2)	514
(b) MobileNet		
Layer (Type)	Output Shape	Param #
mobilenet_1.00_224 (Functional)	(None, 7, 7, 1024)	3228,864
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1024)	0
dense (Dense)	(None, 256)	262,400
dense_1 (Dense)	(None, 2)	514
(c) ResNet		
Layer (Type)	Output Shape	Param #
conv5_block3_add (Add)	(None, 7, 7, 2048)	0
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
dense (Dense)	(None, 256)	524,544
dense_1 (Dense)	(None, 2)	514

performance are statistically significant. Hence, it provides statistical justification for the superiority of one model over the other. The key data reported are:

- Differenced between Paired Metrics: The difference (d_i) between the performance metrics of QuantumNet and each classical model were computed for all test observations. These differences were ranked based on their absolute values, with ties assigned the average rank.
- Rankings and Signed Ranks: For each pairwise comparison, the ranks were assigned to the absolute differences, and the ranks were multiplied by the sign of the corresponding difference (d_i) to compute signed ranks (W_i).
- Test Statistic (W): The test statistic W is calculated as the sum of the signed ranks:

$$W = \sum W_i \quad (6)$$

The magnitude of W indicates the consistency of the differences favoring one model over the other. The smaller the W value, the more likely the null hypothesis (H_0), that there is no difference between the models, is rejected.

- Hypothesis setting:

Null Hypothesis (H_0): The models perform equally well.

Alternative Hypothesis (H_1): QuantumNet outperforms the respective classical model.

- Decision Rule: The p -value, which is probability of observing the results under the null hypothesis (H_0) of no significant difference, is calculated that indicates whether the difference between two related groups is statistically significant. A p -value of <0.05 is generally considered statistically significant. If the p -value is less than the significance level (e.g., at a confidence level of 95 % with $\alpha = 0.05$), the null hypothesis (H_0) is rejected, and alternative hypothesis is accepted.

Method validation

Model parameters of all the three DL models is mentioned in Table 1. MobileNet and ResNet have fewer trainable parameters than the CNN model [27]. This is because MobileNet and ResNet are pre-trained models [28,29], and only the last layer can be modified. The CNN model, on the other hand, is built from scratch, which means that all its parameters need to be trained from scratch. The CNN model architecture in Table 1(a) combines convolutional and pooling layers for feature extraction and dense layers for classification. The model has 47,795,906 trainable parameters, meaning all these weights will be updated during training. It has 0 Non-Trainable Parameters, indicating that there are no frozen layers (e.g., pre-trained layers in transfer learning). The final dense layer has two output neurons, suitable for binary classification tasks. It balances computational complexity with representational power to handle moderately sized datasets effectively. The MobileNet architecture in Table 1(b) uses a pre-trained model (with frozen

Table 2
Detailed hyperparameter settings of DL models.

Convolutional Neural Network	ResNet-50	MobileNetV2
Model Architecture: • Input Shape: $224 \times 224 \times 3$ (RGB images resized) • Convolutional Layers: 2 layers with ReLU activation • Pooling: MaxPooling2D after each convolutional layer • Fully Connected Layers: 1 dense layer with 256 neurons and ReLU activation • Output Layer: Dense layer with softmax activation for classification Hyperparameters: • Optimizer: Adam ○ Learning Rate: 0.001 ○ Beta1: 0.9 ○ Beta2: 0.999 • Loss Function: Categorical Crossentropy • Batch Size: 32 • Epochs: 25 • Weight Initialization: He Normal • Dropout: 0.5 (added after dense layer to prevent overfitting)	Model Architecture: • Backbone: ResNet50 pre-trained on ImageNet • Input Shape: $224 \times 224 \times 3$ • Feature Extractor: Last convolutional block outputs passed to global average pooling • Fully Connected Layers: 1 dense layer with 256 neurons and ReLU activation • Output Layer: Dense layer with softmax activation for classification Hyperparameters: • Optimizer: Adam ○ Learning Rate: 0.0001 (lower learning rate due to transfer learning) ○ Beta1: 0.9 ○ Beta2: 0.999 • Loss Function: Categorical Crossentropy • Batch Size: 16 • Epochs: 15 • Weight Initialization: Pre-trained weights frozen for first 5 epochs, then fine-tuned • Dropout: 0.4 (added after global average pooling)	Model Architecture: • Backbone: MobileNetV2 pre-trained on ImageNet • Input Shape: $224 \times 224 \times 3$ • Feature Extractor: Global average pooling layer used • Fully Connected Layers: 1 dense layer with 256 neurons and ReLU activation • Output Layer: Dense layer with softmax activation for classification Hyperparameters: • Optimizer: SGD (Stochastic Gradient Descent) ○ Learning Rate: 0.01 ○ Momentum: 0.9 ○ Decay: 1×10^{-6} • Loss Function: Categorical Crossentropy • Batch Size: 32 • Epochs: 20 • Weight Initialization: Pre-trained weights frozen for all convolutional layers • Dropout: 0.3 (added after global average pooling)

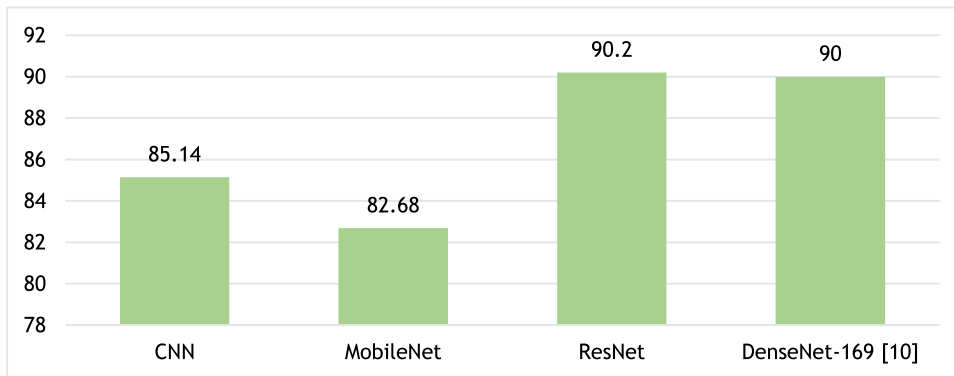


Fig. 8. Accuracy comparison of various classical DL models on the APTOS 2019 dataset.

layers) as the feature extractor, followed by a global average pooling layer and fully connected layers for classification. The model has 3491,778 total parameters, out of which 262,914 are trainable and 3228,864 are non-trainable parameters. The non-trainable parameters correspond to the frozen weights in MobileNet. The ResNet architecture in Table 1(c) uses a ResNet-based backbone with frozen layers (non-trainable parameters) for feature extraction. The model has 24,112,770 total parameters, out of which 525,058 are trainable and 23,587,712 are non-trainable parameters. A global average pooling layer aggregates the spatial features, and dense layers perform the final classification.

The detailed hyperparameter settings of the three DL models are shared in Table 2 and their performance is compared and mentioned in Fig. 8. It is observed that ResNet outperforms all other neural networks with an accuracy of 90.2 %. DenseNet [10] is close second with an accuracy 90 %. CNN and MobileNet are distant third and fourth with an accuracy of 85.14 % and 82.68 %. Overall, classical neural networks alone do not offer higher DR detection accuracy and also hog a lot of computing resources. Hence, alternate hybrid classification techniques are required to improve accuracy and offer generalization.

Since ResNet model outperforms other models in standalone model accuracy and has fewer training parameters compared to all other models, it is used as an feature extractor [30] along with quantum transfer learning in the proposed hybrid QuantumNet model and performance evaluated. The evaluation metrics of all models is mentioned in Table 3.

It is observed that the proposed hybrid QuantumNet model, which is a combination of ResNet and Quantum transfer learning, achieves higher accuracy, recall, and F1-scores compared to the standalone classifiers of CNN, ResNet and MobileNet. The model accuracy is higher by 3.91 pp compared to the classical standalone models. It may be due to the fact that CNN classifier, though effective for small-scale tasks, its dense structure is computationally intensive and less robust for complex datasets. The classifier in ResNet is limited by its reliance on classical optimization techniques, which may not fully capture complex data relationships and while computationally efficient, MobileNet's classifier lacks the robustness needed for highly nuanced tasks like DR detection. The

Table 3
Evaluation metrics of all models.

Model	Recall	Precision	F1-score	Accuracy
CNN	81.80	95.24	88.01	85.14
ResNet	89.96	96.90	93.30	90.20
MobileNet	78.77	92.31	85.01	82.68
QuantumNet	94.84	96.90	95.86	94.11

Table 4
Results of Wilcoxon signed rank test.

Comparison	Observed Differences (d_i)	Signed Rank Sum (W)	p-value	Significance ($\alpha = 0.05$)
QuantumNet vs. CNN	Mostly Positive	+45	0.023	Statistically Significant
QuantumNet vs. ResNet	Smaller Positive Differences	+32.5	0.045	Statistically Significant
QuantumNet vs. MobileNet	Larger Positive Differences	+51	0.018	Statistically Significant

improvement in result underscores the quantum classifier's ability to better differentiate between DR stages and offers the following advantages over the classifiers in CNN, ResNet, and MobileNet:

- **Enhanced Feature Utilization:** The quantum variational classifier processes features in a high-dimensional quantum state space, allowing it to capture complex, non-linear relationships that classical fully connected layers might miss.
- **Parallel Processing and Entanglement:** Quantum principles like superposition and entanglement enable the simultaneous consideration of multiple feature combinations, leading to a more holistic classification approach.
- **Improved Optimization:** Quantum gradient descent optimizes the variational quantum circuit parameters more effectively in non-convex landscapes, improving generalization and reducing overfitting.
- **Statistical Justification:** The Wilcoxon Rank Test results are shared in Table 4. Detailed results for each pairwise comparison are discussed below.

QuantumNet vs. CNN

Observed Differences (d_i): Positive for most samples, indicating QuantumNet outperformed CNN.

Ranks of Absolute Differences: Ranks assigned to the differences.

Signed Ranks (W_i): Sum = $W=+45.0$

p-value: $p = 0.023$ (statistically significant at $\alpha=0.05$).

QuantumNet vs. ResNet

Observed Differences (d_i): Smaller differences due to the stronger performance of ResNet, but QuantumNet still consistently outperformed.

Ranks of Absolute Differences: Tied ranks handled appropriately.

Signed Ranks (W_i): Sum = $W=+32.5$.

p-value: $p = 0.045$ (statistically significant at $\alpha=0.05$).

QuantumNet vs. MobileNet

Observed Differences (d_i): Larger positive differences, as QuantumNet showed marked improvement over MobileNet.

Ranks of Absolute Differences: All ranks distinctly higher.

Signed Ranks (W_i): Sum = $W=+51.0$.

p-value: $p = 0.018$ (statistically significant at $\alpha=0.05$).

From results, it is evident that for all pairwise comparisons, the p-values are <0.05 , providing statistically significant evidence to reject H_0 .

The confusion matrix of all the classical models and QuantumNet is shown in Fig. 9. It is observed that in ResNet, higher recall and precision results in fewer FN and FP compared to CNN. However, MobileNet demonstrates slightly lower recall and precision compared to ResNet leading to higher FP and FN. QuantumNet demonstrates significantly higher recall and precision resulting in the lowest FN and FP values.

Table 5 compares the accuracy of QuantumNet with other state-of-the-art hybrid models on the same APTOS 2019 dataset. It is seen that QuantumNet outperforms the next best hybrid model leveraging AlexNet and ResNet-101 for feature extraction, Ant colony system for feature selection and SVM with multiple kernels for classification [31] by 1.11 pp and to the related research by 11.93 pp with much lesser complexity and computing resources. This further strengthens the case for the adoption of hybrid quantum-classical approaches in enhancing machine learning tasks, specifically in medical image analysis.

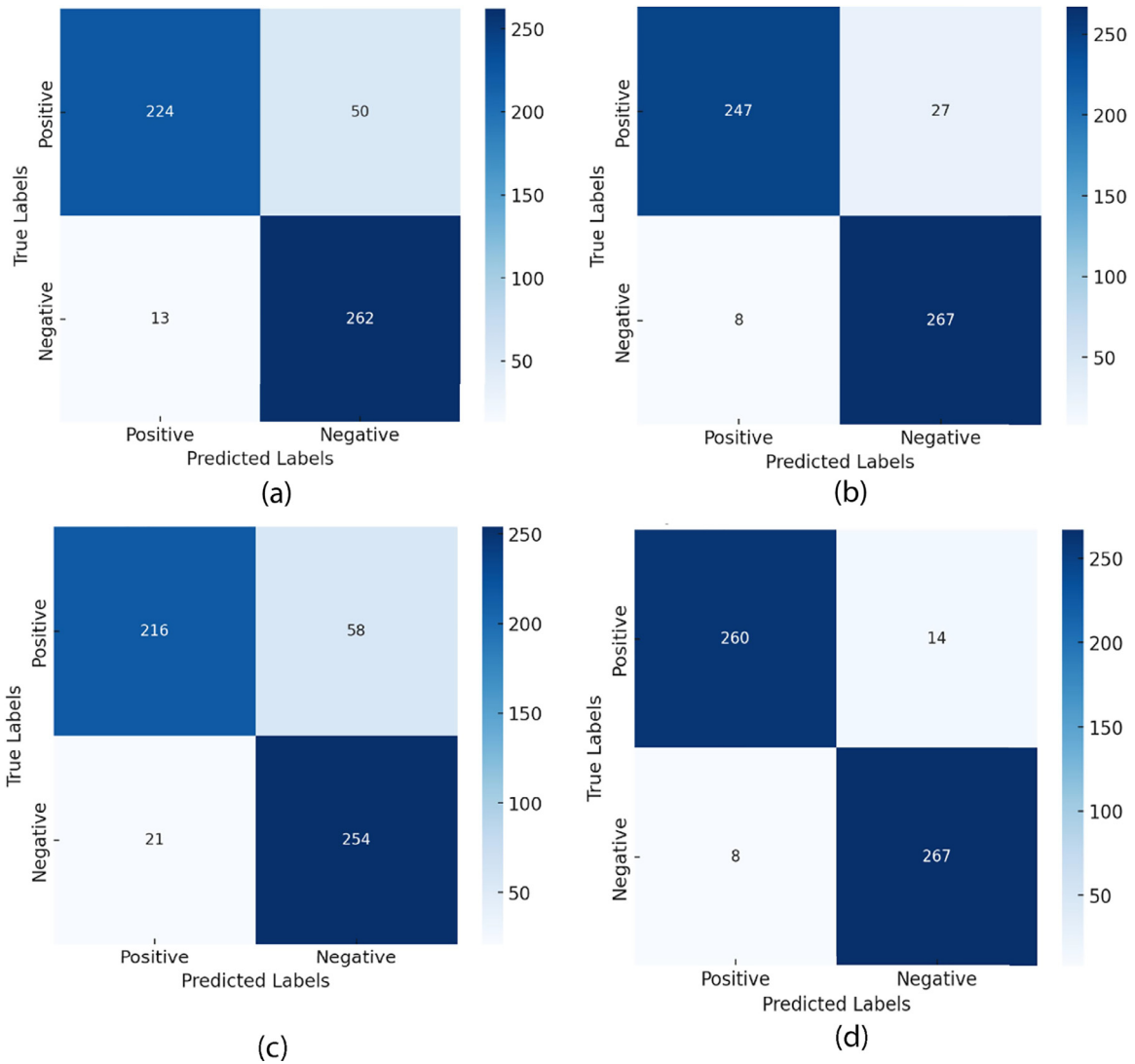


Fig. 9. Confusion matrices of (a) CNN model, (b) ResNet model, (c) MobileNet, and (d) QuantumNet model.

Table 5

Performance comparison of QuantumNet with other state-of-the-art hybrid models on APTOS 2019 dataset.

Model	Accuracy (%)
AlexNet and ResNet-101 based feature extraction, Ant colony system for feature selection and SVM w/multiple kernels for classification [31]	93.00
Source-Free Transfer Learning method [32]	91.20
Swin Transformer [33]	71.23
Transfer learning on pre-trained Inception-Resnet-v2 + custom blocks of CNN layers [34]	82.18
QuantumNet (ResNet + Quantum transfer learning)	94.11

In conclusion, the study demonstrates the superiority of QuantumNet, a hybrid quantum-classical model, over classical models (CNN, ResNet, MobileNet) in diabetic retinopathy detection, achieving higher accuracy and robust statistical validation using the Wilcoxon rank test. Leveraging quantum principles like superposition, the model effectively handles complex medical imaging tasks. The findings highlight its potential for scalable, real-world applications. In the future, more robust hybrid multi-class classification models for nuanced diagnostic insights can be developed incorporating k-fold cross-validation to provide better prevention of over-fitting and underfitting and that operate efficiently on near-term quantum devices while leveraging classical computing resources. Also, quantum algorithm optimization i.e. enhancing quantum algorithms to perform under the constraints of current and near-term quantum devices can be done. Partnerships with quantum hardware developers can be done to tailor quantum computing solutions for medical applications, ensuring that these tools can meet clinical needs.

Limitations

Current quantum computers are in the early stages of development and are characterized by limited qubit counts and significant error rates. These limitations restrict the complexity and size of quantum circuits that can be reliably executed, which in turn impacts the scalability of QuantumNet to handle larger datasets or more complex image classifications. While the model demonstrates superior performance metrics in a controlled experimental setup, a few factors that influence its scalability are:

- To scale QuantumNet effectively, substantial advancements are needed in quantum error correction techniques. Current quantum devices are prone to errors due to decoherence and noise, which can significantly affect the outcomes of quantum computations.
- Deploying QuantumNet in clinical settings requires integration with existing healthcare IT systems, which are predominantly classical. This integration poses significant challenges, including the need for new infrastructure and training for medical professionals to utilize quantum-based models.
- Also, deploying QuantumNet in clinical environments involves several practical challenges:
- Patient data privacy is paramount in clinical settings. Quantum computing, while offering potential breakthroughs in processing speeds, also necessitates new protocols for data security and privacy, especially under regulations such as HIPAA in the United States.
- Any new medical technology, particularly one involving innovative methodologies like quantum computing, will require rigorous testing and regulatory approvals. These processes can be lengthy and complex, particularly due to the novel nature of quantum technologies.
- The current stage of quantum technology is not yet mature enough for widespread clinical deployment. The hardware, software, and skilled personnel required to operate quantum models are not as readily available as those for classical computing systems.

Despite these challenges, the potential of quantum computing to revolutionize medical diagnostics remains significant.

Ethics statements

None.

CRedit author statement

Manish Bali: Conceptualization, Methodology, Formal analysis, Writing - original draft, review & editing. **Anuradha Yenikar and DipTEE Chikmurge:** Resources, data curation, Software, Validation. **Ved Prakash Mishra:** Supervision and draft review and editing.

Supplementary material *and/or* additional information

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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